

4) Tree Topology :-

a. Command :-

- i. `sudo su`
- ii. `mn --topo tree,3`

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Sep 24 11:58
root@sagnik-bachhar-VirtualBox: /home/sagnik-bachhar — sudo su

sagnik-bachhar@sagnik-bachhar-VirtualBox:~$ sudo su
[sudo: authenticate] Password:
root@sagnik-bachhar-VirtualBox:/home/sagnik-bachhar# mn --topo tree,3
*** No default OpenFlow controller found for default switch!
*** Falling back to OVS Bridge
*** Creating network
*** Adding controller
*** Adding hosts:
h1 h2 h3 h4 h5 h6 h7 h8
*** Adding switches:
s1 s2 s3 s4 s5 s6 s7
*** Adding links:
(s1, s2) (s1, s5) (s2, s3) (s2, s4) (s3, h1) (s3, h2) (s4, h3) (s4, h4) (s5, s6) (s5, s7) (s6, h5) (s6, h6) (s7, h7) (s7, h8)
*** Configuring hosts
h1 h2 h3 h4 h5 h6 h7 h8
*** Starting controller

*** Starting 7 switches
s1 s2 s3 s4 s5 s6 s7 ...
*** Starting CLI:
mininet> nodes
available nodes are:
h1 h2 h3 h4 h5 h6 h7 h8 s1 s2 s3 s4 s5 s6 s7
mininet> links
s1-eth1<->s2-eth3 (OK OK)
s1-eth2<->s5-eth3 (OK OK)
s2-eth1<->s3-eth3 (OK OK)
s2-eth2<->s4-eth3 (OK OK)
s3-eth1<->h1-eth0 (OK OK)
s3-eth2<->h2-eth0 (OK OK)
s4-eth1<->h3-eth0 (OK OK)
s4-eth2<->h4-eth0 (OK OK)
s5-eth1<->s6-eth3 (OK OK)
s5-eth2<->s7-eth3 (OK OK)
s6-eth1<->h5-eth0 (OK OK)
s6-eth2<->h6-eth0 (OK OK)
s7-eth1<->h7-eth0 (OK OK)
s7-eth2<->h8-eth0 (OK OK)
```

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Sep 24 11:58
root@sagnik-bachhar-VirtualBox: /home/sagnik-bachhar — sudo su

mininet> links
s1-eth1<->s2-eth3 (OK OK)
s1-eth2<->s5-eth3 (OK OK)
s2-eth1<->s3-eth3 (OK OK)
s2-eth2<->s4-eth3 (OK OK)
s3-eth1<->h1-eth0 (OK OK)
s3-eth2<->h2-eth0 (OK OK)
s4-eth1<->h3-eth0 (OK OK)
s4-eth2<->h4-eth0 (OK OK)
s5-eth1<->s6-eth3 (OK OK)
s5-eth2<->s7-eth3 (OK OK)
s6-eth1<->h5-eth0 (OK OK)
s6-eth2<->h6-eth0 (OK OK)
s7-eth1<->h7-eth0 (OK OK)
s7-eth2<->h8-eth0 (OK OK)
mininet> net
h1 h1-eth0:s3-eth1
h2 h2-eth0:s3-eth2
h3 h3-eth0:s4-eth1
h4 h4-eth0:s4-eth2
h5 h5-eth0:s6-eth1
h6 h6-eth0:s6-eth2
h7 h7-eth0:s7-eth1
h8 h8-eth0:s7-eth2
s1 lo: s1-eth1:s2-eth3 s1-eth2:s5-eth3
s2 lo: s2-eth1:s3-eth3 s2-eth2:s4-eth3 s2-eth3:s1-eth1
s3 lo: s3-eth1:h1-eth0 s3-eth2:h2-eth0 s3-eth3:s2-eth1
s4 lo: s4-eth1:h3-eth0 s4-eth2:h4-eth0 s4-eth3:s2-eth2
s5 lo: s5-eth1:s6-eth3 s5-eth2:s7-eth3 s5-eth3:s1-eth2
s6 lo: s6-eth1:h5-eth0 s6-eth2:h6-eth0 s6-eth3:s5-eth1
s7 lo: s7-eth1:h7-eth0 s7-eth2:h8-eth0 s7-eth3:s5-eth2
```

```
Sep 24 11:58
root@sagnik-bachhar-VirtualBox: /home/sagnik-bachhar — sudo su

sagnik-bachhar@sagnik-ba | root@sagnik-bachhar-Virtu | root@sagnik-bachhar-Virtu | root@sagnik-bachhar-V | x | root@sagnik-bachhar-Virtu | root@sagnik-bachhar-Virtu

s1 lo: s1-eth1:s2-eth3 s1-eth2:s5-eth3
s2 lo: s2-eth1:s3-eth3 s2-eth2:s4-eth3 s2-eth3:s1-eth1
s3 lo: s3-eth1:h1-eth0 s3-eth2:h2-eth0 s3-eth3:s2-eth1
s4 lo: s4-eth1:h3-eth0 s4-eth2:h4-eth0 s4-eth3:s2-eth2
s5 lo: s5-eth1:s6-eth3 s5-eth2:s7-eth3 s5-eth3:s1-eth2
s6 lo: s6-eth1:h5-eth0 s6-eth2:h6-eth0 s6-eth3:s5-eth1
s7 lo: s7-eth1:h7-eth0 s7-eth2:h8-eth0 s7-eth3:s5-eth2
mininet> dump
<Host h1: h1-eth0:10.0.0.1 pid=7218>
<Host h2: h2-eth0:10.0.0.2 pid=7220>
<Host h3: h3-eth0:10.0.0.3 pid=7222>
<Host h4: h4-eth0:10.0.0.4 pid=7224>
<Host h5: h5-eth0:10.0.0.5 pid=7226>
<Host h6: h6-eth0:10.0.0.6 pid=7228>
<Host h7: h7-eth0:10.0.0.7 pid=7230>
<Host h8: h8-eth0:10.0.0.8 pid=7232>
<OVSBridge s1: lo:127.0.0.1,s1-eth1:None,s1-eth2:None pid=7237>
<OVSBridge s2: lo:127.0.0.1,s2-eth1:None,s2-eth2:None,s2-eth3:None pid=7240>
<OVSBridge s3: lo:127.0.0.1,s3-eth1:None,s3-eth2:None,s3-eth3:None pid=7243>
<OVSBridge s4: lo:127.0.0.1,s4-eth1:None,s4-eth2:None,s4-eth3:None pid=7246>
<OVSBridge s5: lo:127.0.0.1,s5-eth1:None,s5-eth2:None,s5-eth3:None pid=7249>
<OVSBridge s6: lo:127.0.0.1,s6-eth1:None,s6-eth2:None,s6-eth3:None pid=7252>
<OVSBridge s7: lo:127.0.0.1,s7-eth1:None,s7-eth2:None,s7-eth3:None pid=7255>
mininet> pingall
*** Ping: testing ping reachability
h1 -> h2 h3 h4 h5 h6 h7 h8
h2 -> h1 h3 h4 h5 h6 h7 h8
h3 -> h1 h2 h4 h5 h6 h7 h8
h4 -> h1 h2 h3 h5 h6 h7 h8
h5 -> h1 h2 h3 h4 h6 h7 h8
h6 -> h1 h2 h3 h4 h5 h7 h8
```

```
Sep 24 11:58
root@sagnik-bachhar-VirtualBox: /home/sagnik-bachhar — sudo su

sagnik-bachhar@sagnik-ba | root@sagnik-bachhar-Virtu | root@sagnik-bachhar-Virtu | root@sagnik-bachhar-V | x | root@sagnik-bachhar-Virtu | root@sagnik-bachhar-Virtu

<OVSBridge s5: lo:127.0.0.1,s5-eth1:None,s5-eth2:None,s5-eth3:None pid=7249>
<OVSBridge s6: lo:127.0.0.1,s6-eth1:None,s6-eth2:None,s6-eth3:None pid=7252>
<OVSBridge s7: lo:127.0.0.1,s7-eth1:None,s7-eth2:None,s7-eth3:None pid=7255>
mininet> pingall
*** Ping: testing ping reachability
h1 -> h2 h3 h4 h5 h6 h7 h8
h2 -> h1 h3 h4 h5 h6 h7 h8
h3 -> h1 h2 h4 h5 h6 h7 h8
h4 -> h1 h2 h3 h5 h6 h7 h8
h5 -> h1 h2 h3 h4 h6 h7 h8
h6 -> h1 h2 h3 h4 h5 h7 h8
h7 -> h1 h2 h3 h4 h5 h6 h8
h8 -> h1 h2 h3 h4 h5 h6 h7
*** Results: 0% dropped (56/56 received)
mininet>
Interrupt
mininet>
Interrupt
mininet> exit
*** Stopping 0 controllers

*** Stopping 14 links
.....
*** Stopping 7 switches
s1 s2 s3 s4 s5 s6 s7
*** Stopping 8 hosts
h1 h2 h3 h4 h5 h6 h7 h8
*** Done
completed in 180.869 seconds
root@sagnik-bachhar-VirtualBox: /home/sagnik-bachhar#
```

b. Command :-

iii. sudo su

iv. mn --topo tree,depth=2,fanout=2

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Sep 24 11:59
root@sagnik-bachhar-VirtualBox: /home/sagnik-bachhar — sudo su

sagnik-bachhar@sagnik-bachhar-VirtualBox:~$ sudo su
[sudo: authenticate] Password:
root@sagnik-bachhar-VirtualBox:/home/sagnik-bachhar# mn --topo tree,depth=2,fanout=2
*** No default OpenFlow controller found for default switch!
*** Falling back to OVS Bridge
*** Creating network
*** Adding controller
*** Adding hosts:
h1 h2 h3 h4
*** Adding switches:
s1 s2 s3
*** Adding links:
(s1, s2) (s1, s3) (s2, h1) (s2, h2) (s3, h3) (s3, h4)
*** Configuring hosts
h1 h2 h3 h4
*** Starting controller

*** Starting 3 switches
s1 s2 s3 ...
*** Starting CLI:
mininet> nodes
available nodes are:
h1 h2 h3 h4 s1 s2 s3
mininet> links
s1-eth1<->s2-eth3 (OK OK)
s1-eth2<->s3-eth3 (OK OK)
s2-eth1<->h1-eth0 (OK OK)
s2-eth2<->h2-eth0 (OK OK)
s3-eth1<->h3-eth0 (OK OK)
s3-eth2<->h4-eth0 (OK OK)
mininet> net
```

```
Sep 24 11:59
root@sagnik-bachhar-VirtualBox: /home/sagnik-bachhar — sudo su

sagnik-bachhar@sagnik-bachhar-VirtualBox:~$ sudo su
[sudo: authenticate] Password:
root@sagnik-bachhar-VirtualBox:/home/sagnik-bachhar# mn --topo tree,depth=2,fanout=2
*** No default OpenFlow controller found for default switch!
*** Falling back to OVS Bridge
*** Creating network
*** Adding controller
*** Adding hosts:
h1 h2 h3 h4
*** Adding switches:
s1 s2 s3
*** Adding links:
(s1, s2) (s1, s3) (s2, h1) (s2, h2) (s3, h3) (s3, h4)
*** Configuring hosts
h1 h2 h3 h4
*** Starting controller

*** Starting 3 switches
s1 s2 s3 ...
*** Starting CLI:
mininet> nodes
available nodes are:
h1 h2 h3 h4 s1 s2 s3
mininet> links
s1-eth1<->s2-eth3 (OK OK)
s1-eth2<->s3-eth3 (OK OK)
s2-eth1<->h1-eth0 (OK OK)
s2-eth2<->h2-eth0 (OK OK)
s3-eth1<->h3-eth0 (OK OK)
s3-eth2<->h4-eth0 (OK OK)
mininet> net
mininet> net
h1 h1-eth0:s2-eth1
h2 h2-eth0:s2-eth2
h3 h3-eth0:s3-eth1
h4 h4-eth0:s3-eth2
s1 lo: s1-eth1:s2-eth3 s1-eth2:s3-eth3
s2 lo: s2-eth1:h1-eth0 s2-eth2:h2-eth0 s2-eth3:s1-eth1
s3 lo: s3-eth1:h3-eth0 s3-eth2:h4-eth0 s3-eth3:s1-eth2
mininet> dump
<Host h1: h1-eth0:10.0.0.1 pid=7702>
<Host h2: h2-eth0:10.0.0.2 pid=7704>
<Host h3: h3-eth0:10.0.0.3 pid=7706>
<Host h4: h4-eth0:10.0.0.4 pid=7708>
<OVSBridge s1: lo:127.0.0.1,s1-eth1:None,s1-eth2:None pid=7713>
<OVSBridge s2: lo:127.0.0.1,s2-eth1:None,s2-eth2:None,s2-eth3:None pid=7716>
<OVSBridge s3: lo:127.0.0.1,s3-eth1:None,s3-eth2:None,s3-eth3:None pid=7719>
mininet> pingall
*** Ping: testing ping reachability
h1 -> h2 h3 h4
h2 -> h1 h3 h4
h3 -> h1 h2 h4
h4 -> h1 h2 h3
*** Results: 0% dropped (12/12 received)
mininet> h1 ping -c 4 h4
PING 10.0.0.4 (10.0.0.4) 56(84) bytes of data.
64 bytes from 10.0.0.4: icmp_seq=1 ttl=64 time=1.12 ms
64 bytes from 10.0.0.4: icmp_seq=2 ttl=64 time=0.055 ms
64 bytes from 10.0.0.4: icmp_seq=3 ttl=64 time=0.055 ms
```

```
Sep 24 11:59
root@sagnik-bachhar-VirtualBox: /home/sagnik-bachhar — sudo su

sagnik-bachhar@sagnik-ba | root@sagnik-bachhar-Virtu | root@sagnik-bachhar-Virtu | root@sagnik-bachhar-Virtu | root@sagnik-bachhar-V | x | root@sagnik-bachhar-Virtu

h4 h4-eth0:s3-eth2
s1 lo: s1-eth1:s2-eth3 s1-eth2:s3-eth3
s2 lo: s2-eth1:h1-eth0 s2-eth2:h2-eth0 s2-eth3:s1-eth1
s3 lo: s3-eth1:h3-eth0 s3-eth2:h4-eth0 s3-eth3:s1-eth2
mininet> dump
<Host h1: h1-eth0:10.0.0.1 pid=7702>
<Host h2: h2-eth0:10.0.0.2 pid=7704>
<Host h3: h3-eth0:10.0.0.3 pid=7706>
<Host h4: h4-eth0:10.0.0.4 pid=7708>
<OVSBridge s1: lo:127.0.0.1,s1-eth1:None,s1-eth2:None pid=7713>
<OVSBridge s2: lo:127.0.0.1,s2-eth1:None,s2-eth2:None,s2-eth3:None pid=7716>
<OVSBridge s3: lo:127.0.0.1,s3-eth1:None,s3-eth2:None,s3-eth3:None pid=7719>
mininet> pingall
*** Ping: testing ping reachability
h1 -> h2 h3 h4
h2 -> h1 h3 h4
h3 -> h1 h2 h4
h4 -> h1 h2 h3
*** Results: 0% dropped (12/12 received)
mininet> h1 ping -c 4 h4
PING 10.0.0.4 (10.0.0.4) 56(84) bytes of data.
64 bytes from 10.0.0.4: icmp_seq=1 ttl=64 time=1.12 ms
64 bytes from 10.0.0.4: icmp_seq=2 ttl=64 time=0.055 ms
64 bytes from 10.0.0.4: icmp_seq=3 ttl=64 time=0.055 ms
64 bytes from 10.0.0.4: icmp_seq=4 ttl=64 time=0.041 ms

--- 10.0.0.4 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3078ms
rtt min/avg/max/mdev = 0.041/0.318/1.122/0.464 ms
mininet>
```